

Section 1: Data Understanding Tasks:

1. Load dataset
2. Display: Shape and Columns
3. Check: Missing values and Data types

```
1 import pandas as pd # imported pandas and stored it in the variab
2 df = pd.read_csv('Mall_Customers.csv') # read the csv file
3 df # get the data
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows x 5 columns

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
1 df.shape # the number of rows and columns
```

```
(200, 5)
```

```
1 df.columns # name of columns
```

```
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
      'Spending Score (1-100)'],
      dtype='object')
```

```
1 df.info() # To get the data types
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
#   ...
```

```

-----
0  CustomerID      200 non-null    int64
1  Gender          200 non-null    object
2  Age             200 non-null    int64
3  Annual Income (k$) 200 non-null    int64
4  Spending Score (1-100) 200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB

```

```
1 df.isnull() # to check for the missing values in the data set
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
195	False	False	False	False	False
196	False	False	False	False	False
197	False	False	False	False	False
198	False	False	False	False	False
199	False	False	False	False	False

200 rows × 5 columns

```
1 df.isnull().sum() # To count the number of missing values from each column
```

	0
CustomerID	0
Gender	0
Age	0
Annual Income (k\$)	0
Spending Score (1-100)	0

dtype: int64

```
1 df.isnull().sum().sum() # Count all missing values from the rows
```

np.int64(0)

Questions:

1. What type of dataset it is?
2. Which features are numerical/categorical?

Ans 1. Type of Dataset: It is a structured tabular dataset that contains customer information.

Ans2. Numerical features are:

1. CustomerID
2. Age
3. Annual Income(k\$)
4. Spending Score (1-100)

Categorical features are:

1. Gender

Section 2: Descriptive Analysis Tasks: Compute the mean, median, and standard deviation for:

1. age
2. annual income
3. spending score

```
1 df['Age'].mean() # to get the mean
```

```
np.float64(38.85)
```

```
1 df['Age'].median() # to get the median
```

```
36.0
```

```
1 df['Age'].std() # to get the standard deviation
```

```
13.969007331558883
```

```
1 df['Annual Income (k$)'].mean() # to get the mean
```

```
np.float64(60.56)
```

```
1 df['Annual Income (k$)'].median() # to get the median
```

```
61.5
```

```
1 df['Annual Income (k$)'].std() # to get the standard deviation
```

```
26.264721165271254
```

```
1 df['Spending Score (1-100)'].mean() # to get the mean
```

```
np.float64(50.2)
```

```
1 df['Spending Score (1-100)'].median() # to get the median
```

```
50.0
```

```
1 df['Spending Score (1-100)'].std() # to get the standard deviation
```

```
25.823521668370162
```

Questions:

1. Which feature shows highest variability?
2. Are customers similar or diverse?

Ans 1. Highest variability means we look which feature has the highest standard deviation. Which is 'Annual Income (k\$)'.

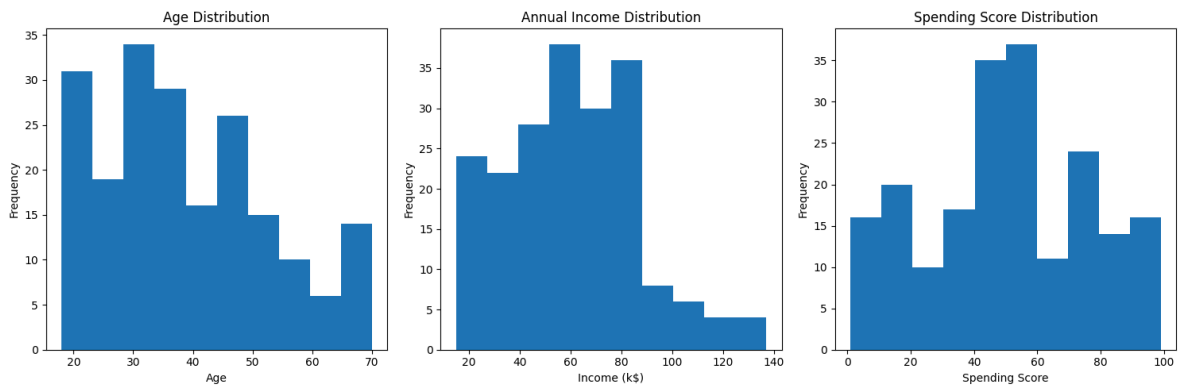
Ans 2. Customers are diverse, as the dataset shows there is significant variability in annual income and spending scores. Which indicates different customer profiles and purchasing behaviors.

Section 3: Visualization Tasks:

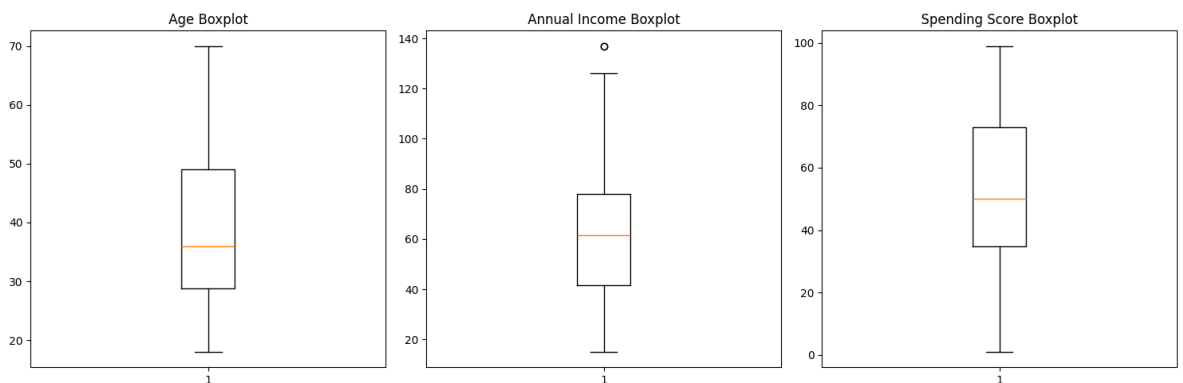
1. Histogram: (Age, Income, Spending)
2. Boxplot: (Detect outliers)

```
1 import matplotlib.pyplot as plt # Import Matplotlib for data visu
2 plt.figure(figsize=(15,5)) # Create a figure of size 15x5 inches
3
4 plt.subplot(1,3,1) # Create subplot 1 (1 row, 3 columns, position
5 plt.hist(df['Age'], bins=10) # Plot histogram for Age with 10 bin
6 plt.title('Age Distribution') # Set title for Age histogram
7 plt.xlabel('Age') # Label x-axis
8 plt.ylabel('Frequency') # Label y-axis
9
10 plt.subplot(1,3,2) # Create subplot 2 (position 2)
11 plt.hist(df['Annual Income (k$)'], bins=10) # Plot histogram for
12 plt.title('Annual Income Distribution') # Set title for Income hi
13 plt.xlabel('Income (k$)') # Label x-axis
14 plt.ylabel('Frequency') # Label y-axis
15
16 plt.subplot(1,3,3) # Create subplot 3 (position 3)
17 plt.hist(df['Spending Score (1-100)'], bins=10) # Plot histogram
18 plt.title('Spending Score Distribution') # Set title for Spending
19 plt.xlabel('Spending Score') # Label y-axis
20
```

```
21 plt.tight_layout() # Adjust spacing between subplots
22 plt.show() # Display all histograms
```



```
1 import matplotlib.pyplot as plt # Import Matplotlib for visualiza
2
3 plt.figure(figsize=(15,5)) # Create a figure of size 15x5 inches
4
5 plt.subplot(1,3,1) # Create subplot 1
6 plt.boxplot(df['Age']) # Create boxplot for Age
7 plt.title('Age Boxplot') # Set title
8
9 plt.subplot(1,3,2) # Create subplot 2
10 plt.boxplot(df['Annual Income (k$)']) # Create boxplot for Annual
11 plt.title('Annual Income Boxplot') # Set title
12
13 plt.subplot(1,3,3) # Create subplot 3
14 plt.boxplot(df['Spending Score (1-100)']) # Create boxplot for Sp
15 plt.title('Spending Score Boxplot') # Set title
16
17 plt.tight_layout() # Adjust spacing between plots
18 plt.show() # Display all boxplots
```



Questions:

1. Is data skewed?
2. Are there extreme customers?

Ans 1. By looking at the histogram of age distribution it appears to be slightly right - skewed, as there are less customers of higher age. Now, for the annual income distribution it is positively right skewed as there are less customers who earn more. Now for the spending score it appears to be fairly balanced with no strong skewness.

To conclude The dataset shows slight positive skewness, particularly in Annual Income.

Ans 2. Yes. The Annual Income boxplot shows at least one extreme customer with unusually high income compared to the rest of the customers. No significant outliers are observed in Age or Spending Score.

Section 4: Group-based Analysis Tasks:

Compare Male vs Female customers by computing:

1. Average spending score
2. Average income

```
1 df.groupby('Gender')['Spending Score (1-100)'].mean() # Calcul
```

Spending Score (1-100)

Gender

Female	51.526786
--------	-----------

Male	48.511364
------	-----------

dtype: float64

```
1 df.groupby('Gender')['Annual Income (k$)'].mean() # Calcul
```

Annual Income (k\$)

Gender

Female	59.250000
--------	-----------

Male	62.227273
------	-----------

dtype: float64

Questions:

1. Which group spends more ?
2. Is there a visible difference ?

Ans1 . Female customers spend more on average than male customers, with an average spending score of 51.53 compared to 48.51 for males.

Ans2 . $51.53 - 48.51 = 3.02$ There is only a small difference in spending behavior between male and female customers. Female customers have a slightly higher average spending score, but the difference is not very large.

Female Average Income = 59.25 k *MaleAverageIncome* = 62.23k

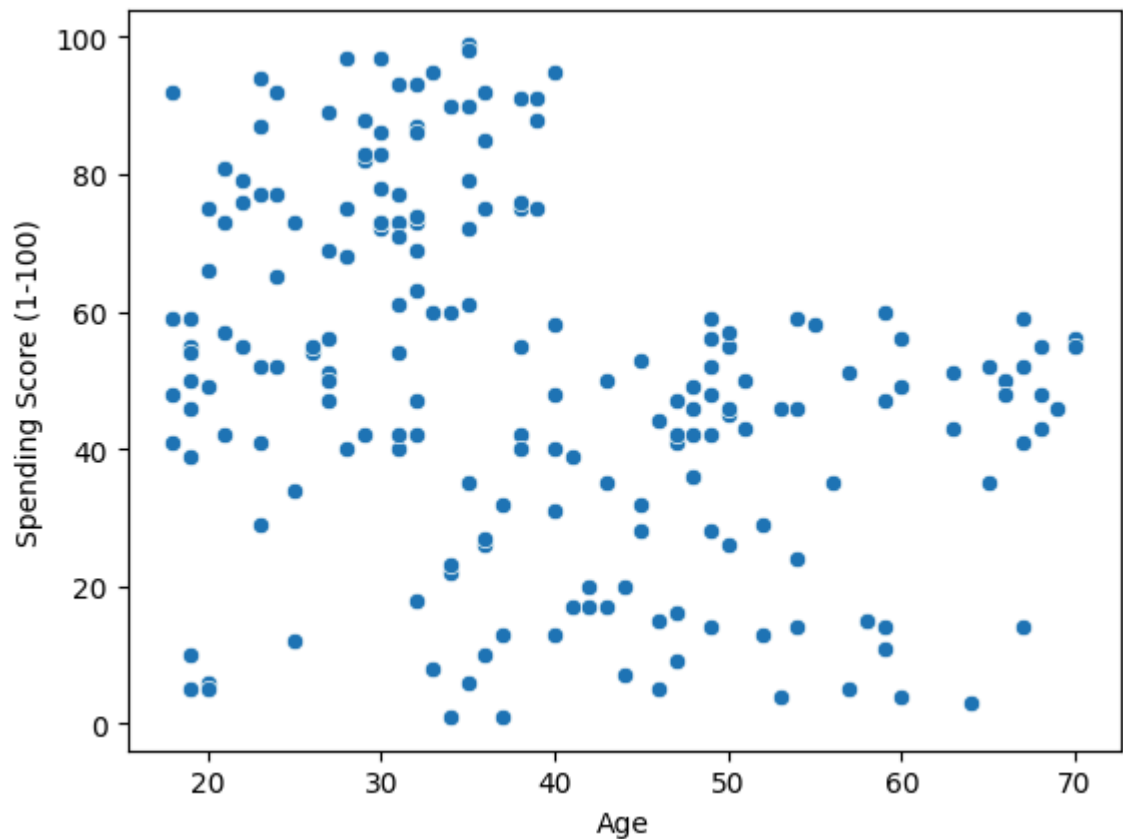
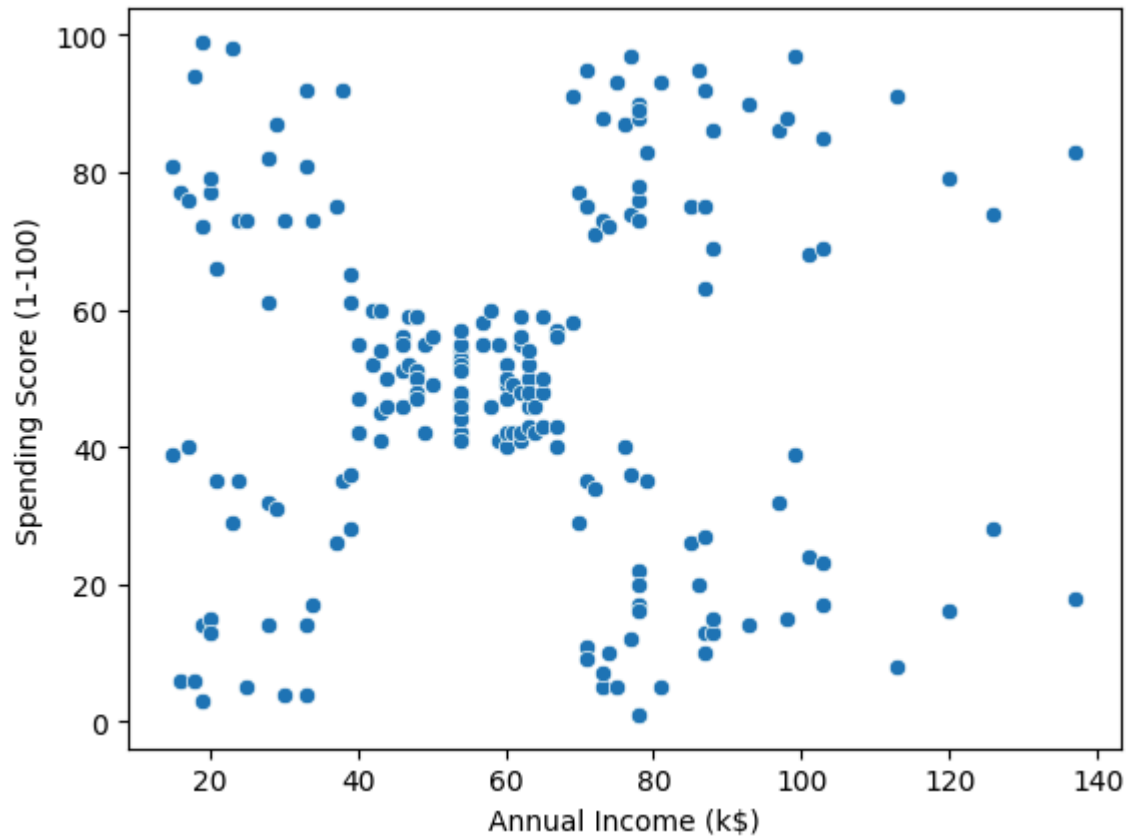
Although males earn slightly more on average, females tend to spend slightly more at the mall.

Section 5: Correlation Analysis Tasks:

Identify correlation between:

1. Income vs Spending score
2. Age vs Spending score

```
1 import seaborn as sns
2 import matplotlib.pyplot as plt
3
4 # Scatter plot: Income vs Spending Score
5 sns.scatterplot(x='Annual Income (k$)', y='Spending Score (1-100)')
6 plt.show()
7
8 # Scatter plot: Age vs Spending Score
9 sns.scatterplot(x='Age', y='Spending Score (1-100)', data=df) #
10 plt.show()
```



Questions:

1. Do richer customers spend more ?
2. Does age affect spending ?

Ans 1. No, higher-income customers do not necessarily spend more. The scatter plot does not show a clear upward trend between Annual Income and Spending Score. Customers with both low and high incomes are distributed across low, medium, and high spending scores. Therefore, income alone is not a strong predictor of spending behavior.

Ans 2. Yes, age appears to influence spending behavior. The scatter plot suggests a negative relationship between Age and Spending Score. Younger customers generally have higher spending scores, while older customers tend to have lower spending scores. This indicates that spending tends to decrease as age increases.

Section 6: Inferntial Statistics(CORE)

Problem Statement: "Do male and female customers spend differently?"

Step 1: Hypothesis H_0 : No difference in spending. H_1 : There is a difference

Step 2: Perform Test Python
`from scipy import stats`
`male = df[df['Gender'] == 'Male']['Spending Score (1-100)']`
`female = df[df['Gender'] == 'Female']['Spending Score (1-100)']`
`t_stat, p_value = stats.ttest_ind(male, female)`
 Step 3: Decision
 ○ If $p < 0.05 \rightarrow$ Reject H_0

```
1 from scipy import stats # Import statistical functions from scipy
2
3 male = df[df['Gender'] == 'Male']['Spending Score (1-100)'] # Fil
4 female = df[df['Gender'] == 'Female']['Spending Score (1-100)'] #
5
6 t_stat, p_value = stats.ttest_ind(male, female) # Perform indepen
7
8 print("T - statistics:", t_stat) # Print t-statistic value
9 print("P - value:", p_value) # Print p-value result of the test
10
11 if p_value < 0.05: # Check if result is statistically significant
12     print("Reject Null Hypothesis") # If p < 0.05, there is signi
13 else:
14     print("Accept Null Hypothesis") # If p >= 0.05, no significan
```

```
T - statistics: -0.8190464150660334
P - value: 0.4137446589852174
Accept Null Hypothesis
```

Questions:

Q1. What does p-value tell you?

Q2. Is the difference statistically significant?

Ans1. The p-value tells you how likely it is to get the observed result (or something more extreme) if the null hypothesis is actually true. It helps you understand whether the difference you see in the data is due to random chance or a real effect.

Ans2. No, the difference is not statistically significant because the p-value (0.4137) is greater than 0.05. This means we do not have enough evidence to reject the null hypothesis, so we conclude that there is no significant difference in spending between male and female customers.

Section 7: Confidence Interval

Tasks:

Calculate 95% CI for the spending score

```

1 import numpy as np # Import NumPy for numerical operations
2 import scipy.stats as stats # Import statistical functions (not u
3
4 # Take the spending score column
5 spending = df['Spending Score (1-100)'] # Extract spending score
6
7 # Sample statistics
8 mean = np.mean(spending) # Calculate mean of spending scores
9 std = np.std(spending, ddof=1) # Calculate sample standard deviat
10 n = len(spending) # Get total number of observations
11
12 # Z value for 95% confidence
13 z = 1.96 # Critical Z-value for 95% confidence interval
14
15 # Standard error
16 se = std / np.sqrt(n) # Compute standard error of the mean
17
18 # Confidence interval
19 lower = mean - z * se # Lower bound of 95% confidence interval
20 upper = mean + z * se # Upper bound of 95% confidence interval
21
22 print("Mean Spending Score:", mean) # Print mean value
23 print("95% Confidence Interval:", (lower, upper)) # Print confide

```

Mean Spending Score: 50.2

95% Confidence Interval: (np.float64(46.621042491978834), np.float64(53.78))

Questions:

Q1: What range do customers fall into?

Q2: How confident are we?

Ans1: The average spending score of customers is 50.2, and the 95% confidence interval is approximately (46.62, 53.78). This means the true average spending score

of all customers is expected to lie between 46.62 and 53.78.

Ans2: We are 95% confident that the true population mean spending score falls within this range. This means that if we repeated this sampling process many times, about 95 out of 100 such intervals would contain the true average spending score.

Section 8: Business Insights

Q1. Which customers should the business target?

Q2. Does income affect spending?

Q3. Should marketing differ by gender?

Q4. What strategy would you suggest?

Ans1. The business should target customers based on their spending behavior rather than gender or income. From the analysis, customers with higher spending scores are the most valuable group, so the business should focus more on high-spending customers identified through clustering.

Ans2. No, income does not clearly affect spending. From the scatter plot, there is no strong relationship between annual income and spending score. Customers with both high and low income show mixed spending behavior.

Ans3. No, marketing should not differ by gender. The t-test shows that there is no statistically significant difference in spending between male and female customers, so gender is not an important factor for targeting.

Ans4. The best strategy is to focus on customer segmentation based on behavior. The business should target high-spending customers with premium offers and use